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Course : Data Analytics
Target Audience : Beginner Students

(1) A Database is an organized collection of structured data stored electronically for efficient storage, retrieval, update, and management.

→ Databases handle massive data volumes generated daily, enabling quick analysis and informed decisions across industries. They ensure data integrity, support scalability for growing operations, and allow multiple users to access information simultaneously without conflicts.

Real world examples

* Banks use relational databases to track transactions, customer accounts, and ensure real-time security for ATMs online banking.

* Social media like Facebook stores user profiles and posts to deliver targeted ads and recommendations.

(2) Data warehouse → Data ~~what~~ warehouse is a centralized repository that stores large volumes of historical, integrated data from multiple sources, optimized specifically for complex queries, reporting and business intelligence analysis.

* It differs from a regular database because databases handle real-time transactional data (OLTP) with frequent updates for daily operations, while data warehouse focus on analytical processing (OLAP) with historical data that's refreshed periodically. Companies use data warehouse to uncover trends across years of data, improve decision-making,

and create a single source of truth for reporting such as sales forecasts or customer behavior analysis.

(3) Data Lake → is a centralized storage system that holds raw, unprocessed data in its native format (structured, semi-structured, or unstructured) at scale, allowing flexible storage without predefined schemas.

* Unlike data warehouses, which requires structured, cleaned data in specific schemas for analysis, data lakes store everything (as-is) for later processing, making them cheaper for big data volumes. Businesses prefer data lakes when dealing with massive, varied data types like IoT sensor data, logs, or multimedia, where they need to store first and analyze later such as in AI experimentation or real-time streaming analytics.

(4) Data Mart → is a focused subset of data warehouse tailored to specific needs of single department or business unit, containing summarized data for targeted analysis.

* It relates to a data warehouse as a smaller, specialized slice often created from the warehouse via ETL processes making queries faster for niche users like marketing or finance teams. Business analysts, sales performance segmentation, without accessing the full enterprise warehouse.

(5) Structured, Semi-structured, Unstructured data

* Structured data is highly organized in fixed formats like tables with rows / columns (e.g. SQL databases with customer IDs, names and addresses).

* Semi-structured has some organization via tags or markers but no rigid schema.

(e.g. JSON or XML) XML files with user profiles including flexible (like hobbies).

• Unstructured data ~~data~~ lacks predefined structure (e.g. emails, videos, or social media posts), comprising 80-90% of enterprise and requiring advanced data tools for analysis.

(6) Data modelling → is the process of creating a visual diagram or blueprint of data structures, relationships and rules to design efficient databases or system.

• Its important because it ensures data integrity, reduces redundancy, supports scalability and prevents errors during development like, mismatched fields causing query failures. Two types: Conceptual models show high-level entities / relationships (e.g. Customer buys product); Logical model adds attributes and keys (e.g. primary keys); Physical models specify storage details like table schemas.

(7) Data Mining → is the process of discovering patterns, correlations, and anomalies in large datasets using statistical, machine learning and database technique. Businesses use it to gain competitive advantages through customer segmentation, predictive analytics, and market basket analysis. Real-world examples include Netflix recommending shows based on viewing patterns and banks detecting fraudulent transactions through unusual behavior detection.

(8) Database Management System (DBMS)
A DBMS is software that enables users to

create, read, update and delete data in a database while ensuring security, integrity and concurrent access. Popular platforms include: MySQL (relational, used for web applications like WordPress); PostgreSQL (advanced SQL features for complex queries in analytics); Oracle (enterprise-scale with high security for banking); MongoDB (NoSQL) document store for flexible, scalable apps like content management.

(a) ETL (Extract, Transform, Load)

→ ETL extracts raw data from multiple sources, transforms it by cleaning, aggregating, and formatting for consistency, then loads it into a target system like a data warehouse. This process is crucial in data analytics because it creates reliable, unified datasets from disparate, messy sources for accurate reporting and analysis.

(10) Data Storage Formats

CSV (Comma-separated values) stores tabular data simply and is ideal for small exports or Excel imports. JSON (Javascript Object Notation) handles nested, hierarchical data for APIs and web apps. Parquet and Avro are columnar/compressed formats optimized for big data analytics. We use Parquet for fast queries in tools like Spark, Avro for schema evolution in streaming pipelines.

(11) Artificial Intelligence (AI)

Artificial intelligence is the simulation of human intelligence in machines to perform tasks like reasoning, learning and problem-solving. Narrow AI (weak AI) handles specific tasks like voice assistance (e.g. Siri). General AI (strong AI)

achieves broad human-level intelligence across any task (still theoretical)

(12) Machine Learning

Machine learning enables computers to learn patterns from data without explicit programming rules. Unlike traditional programming (fixed if-then rules), ML improves automatically with more data. Main types: Supervised (labeled data for prediction / classification), unsupervised (finding hidden patterns / clustering), Reinforcement (learning via rewards / penalties like Go/AlphaGo)

(13) Deep Learning

Deep learning is a subset of ML using multi-layered neural networks to automatically learn hierarchical features from raw data. It relates to ML/AI as an advanced technique, empowered by vast data/compute. Artificial neural networks work by interconnected nodes (neurons) processing inputs through weight layers, adjusting via backpropagation to minimize errors (e.g. image recognition).

(14) Natural Language Processing (NLP)

NLP enables computers to understand, interpret and generate human language. Everyday applications include virtual assistants (Siri) machine translation (Google Translate), and sentiment analysis on social media reviews

(15) Large Language Model (LLM)

A large language model is a deep learning model trained on massive text datasets to generate human-like language. Unlike standard ML models, LLMs handle context

nance, and long sequence at scale. Example: GPT-4 (chat bots / conversation), BERT (search ranking / understanding)

(16) Generative AI

Generative AI creates new content (text, image, code) by learning from training data and sampling variations, i.e. works models like GANs or transformers predicting next tokens/elements. Tools ChatGPT (text generation), DALL-E (images from descriptors), GitHub Copilot (code suggestions).

(17) Training Dataset

→ A training dataset is the labeled or raw data used to teach AI models patterns and behaviors. Quality matters because 'garbage-in-garbage-out' leads to poor accuracy; biased data causes unfair outcomes like discriminatory loan approval or facial recognition errors for certain ethnicities.

(18) Computer Vision

Computer vision lets machines interpret visual data from images/videos using techniques like edge detection and CNNs. Machines "see" by breaking images into pixels/features processed through neural networks. Use cases: Autonomous vehicles (obstacle detection), medical imaging (tumor identification).

(19) API

An API (Application Programming Interface) is a set of rules allowing software to communicate and share data/functions. Analogy: restaurant menu; waiter (API) takes your order (request) to kitchen (service) and delivers.

food (response). APIs enable integration in modern systems like embedding Google Maps in apps.

(20) API key

An API key is a unique string identifying/authenticating users for API access, controlling usage quotas. Store it securely in environment variables or vaults, never in code, exposure risks unauthorized access, high bills, or data breaches.

(21) REST API

REST (Representational State Transfer) is an architectural style for stateless web APIs using standard HTTP. Main methods: GET (retrieve data), POST (create new), PUT (update/replace), DELETE (remove).

(22) JSON

JSON is a lightweight, human-readable format for structured data exchange in APIs, it's preferred for simplicity and language independence. Example: {"name": "Alice", "age": 28, "city": "Pretoria"}

(23) Prompt Engineering

Prompt Engineering crafts precise inputs to optimize AI outputs. Essential for tools like Chat GPT to reduce hallucinations/get targeted results. Poor: "Tell me about dogs." Better: "List 5 breeds of dogs suitable for apartments, with pros/cons in bullet points." Poor: "write story"; Better: "write 300-word sci-fi story about AI in 2500, first-person perspective."

(24) MAP Function

MAP transforms each item in an array using a function, returning a new array. Python example: Numbers = [1, 2, 3]; doubled = list(map(lambda x: x * 2, numbers)) → [2, 4, 6] used for batch data processing.

(25) Cloud Computing

Cloud computing delivers on-demand IT resources over internet. Models: IaaS (Infrastructure, e.g. AWS EC2 virtual machines), PaaS (Platform, e.g. Google App Engine for app deployment), SaaS (Software, e.g. Microsoft 365 for email/docs).

(26) Version Control

A Version Control System tracks changes in code/files collaboratively. Git is distributed enabling branching/merging. Commit saves a snapshot with message; branch creates parallel version for features without affecting main code.

(27) Jupyter Notebook

Jupyter Notebook is an interactive web environment mixing code, visuals, and text. Primary for Python in data analysis/science due to real-time execution, charts and reproducibility.

(28) Python

Python is a versatile, ~~easy~~ readable language dominating data analysis/AI for simplicity and libraries. Popular libraries: Pandas (data manipulation), NumPy (numerical arrays), Matplotlib/Seaborn (visualizations), Scikit-learn (machine learning algorithms).

(29) Data Privacy

Data privacy protects personal information from unauthorized access (use). GDPR (General Data Protection Regulation) is EU law mandating consent, transparency, and rights data deletion, applies to any company handling EU residents' data globally.

(30) Data Ethics

Data ethics governs moral use of data (AI). Unethical example: Cambridge Analytica is unauthorized voter profiling (privacy violation) manipulation, biased COMPAS recidivism algorithm (racial disparities) deepfake video spreading election misinformation (trust erosion).

(31) Data Governance

Data governance is the overall management of data availability, usability, integrity and security through policies, standards and processes. Key components include data quality rules, metadata management, access control, stewardship roles, and compliance monitoring to ensuring trusted data for decision-making.

(32) Bias AI

AI bias occurs when models produce unfair or inaccurate outcomes due to skewed training data or algorithms that reflect human prejudices. Example: facial recognition system performing poorly on darker skin tones, leading to higher rates of minorities and consequences like wrongful arrests or discrimination.

(33) Data Roles Comparison

Role	Main Focus	Key Skills	Common Tools
Data Analyst	Business insights from data	SQL, Excel, Visualization, Stats	Tableau, Power BI, Google Sheets
Data Scientist	Predictive modeling ML	Python/R, ML algorithms, Stats	Jupyter, Scikit-learn, TensorFlow
Data Engineer	Data pipeline / infrastructure	ETL, Cloud, data databases, programming	Airflow, Spark, AWS / GCP, Docker
AI Engineer	Deploying AI Systems	Deep learning MLOps, APIs	PyTorch, Kubernetes, FastAPI

(34) Business Intelligence (BI)

Business Intelligence. Involves tools and strategies to analyze data for actionable business insights through dashboards and reports. Popular tools: Power BI (Microsoft interactive dashboards from multiple sources); Tableau (visual analytics with drag and drop for storytelling). They help by turning raw data into visual for quick decisions like sales trends or performance KPIs.

(35) Most Interesting Concept

(Personal: ~~Generative~~ AI Example: "Generative AI surprised me most because it creates original art / code from text prompts revolutionizing creative work").

(36) Most Difficult Concept

Deep learning was harder due to neural network math. Used AWS ~~explaining~~ and YouTube visuals to understand layers / backpropagation.

(37) Data / AI Career Impact

In public health AI will analyze HIV
recurrence patterns for better intervention,
daily life via personalized health apps
adherence.

References

- (1) Causera. (2022). Data Warehouse vs Database.
<https://www.causera.org/articles/data-warehouse-vs-database/>
- (2) TechTarget. (2024) What is database (DB)?
<https://www.techtarget.com/searchdatamanagement/definition/database/>
- (3) GeeksForGeeks. (2019) Difference between Database and Data Warehouse
<https://www.geeksforgeeks.org/databases/difference-between-database-system-and-database-warehouse/>
- (4) Imperia. (2023). Data Warehouse Benefits. <https://www.imperia.com/learn/data-security/data-warehouse/>
- (5) Tulane University. (2024). How is Data Mining Used in Business? <https://online.sse.tulane.edu/articles/data-mining>
- (6) Google. (2025) Generative AI Overview
<https://www.deepmind.google/technologies/generative-ai/>
- (7) IBM. (2024) What is Computer Vision
<https://www.ibm.com/topics/computer-vision>
- (8) Dev.to. (2025) Real world DBMS Examples, Explained with the use cases. <https://dev.to/kishubhapt/real-world-dbms-examples-explained-with-use-cases-2amj>
- (9) Pipe drive. (2025) 8 Top Database Management

System. (DBMS) <https://www.pipedrive.com/en/blog/database-management-systems>

(10) Microsoft Azure. (2025) Training Data Best Practices. <https://azure.microsoft.com/en/resources/ai-training-data/>